WQBEL BIOLOGIST RECOMMENDATIONS

FACT/DECISION SHEET

Cadillac Waste Water Treatment Plant (WWTP) - MI0020257 - REISSUANCE

PREPARED BY: GLEN SCHMITT

1/30/2018

Background

The City of Cadillac submitted a National Pollutant Discharge Elimination System (NPDES) permit application on April 3, 2017 for reissuance of the Cadillac WWTP located in Cadillac, Michigan, Wexford County. The current NPDES permit was issued on September 27, 2013, modified on December 10, 2015, effective on January 1, 2016, and expired on October 1, 2017, but has been extended due to submittal of a timely permit reissuance application. The current NPDES permit authorizes to continuously discharge a maximum of 3.2 million gallons per day (MGD) of treated municipal wastewater and discharge it through Monitoring Point 001A through Outfall 001 to the Clam River. The Clam River has a 95 percent exceedance, harmonic mean, and 90dQ10 low flows of 2.9, 14, and 3.6 cubic feet per second (cfs), respectively (Low Flow File No. 8970). The Clam River is a warm water stream and is not known as a public water supply source.

The hardness value used for this review was 75 mg/l. This value was measured from the Clam River at Smith Street Bridge (44.26330 N, -85.40099 W) on 10/14/2014 during a Point Source Monitoring Survey.

Sources

Sources of information used for this review to develop the water quality-based effluent limits (WQBELs) include the facility's NPDES permit reissuance application, the current NPDES permit, the Permits Section Toxics facility file, discharge monitoring report data from June 2013 through October 2017 as this is the available final effluent data since the previous WQBEL review, a Point Source Monitoring Survey conducted on October 14, 2014, and United States Geological Survey topographical maps. Based upon our review of this information, we have the following recommendations:

Recommendations

1) The 2016 Water Quality and Pollution Control in Michigan Sections 303(d), 305(b), and 314 Integrated Report indicates the Clam River (Assessment Unit ID (AUID) 040601020303-01) is Not Supporting Water Quality Standards (WQS) (Category 4a Waterbody) for the Fish Consumption designated use due to polychlorinated biphenyls (PCBs) in the water column. All other designated uses are listed as Fully Supporting or Not Assessed. A statewide Total Maximum Daily Load (TMDL) for PCBs was approved by the United States Environmental Protection Agency (USEPA) on September 26, 2017. Several of the downstream AUIDs also have similar nonattainment issues.

Outfall 001

2) The current NPDES permit requires daily flow reporting. The Outfall 001 final effluent flow from June 2013 through October 2017 ranged from 1.14 to 3.61 MGD, averaging 2.1 MGD as compared to the design flow of 3.2 MGD. There was one day during this time period when the final effluent flow was greater than 3.2 MGD. Based on this information, we recommend the daily monitoring requirements for flow to be retained in the draft NPDES permit.

3) The current NPDES permit includes daily minimum and daily maximum final effluent limitations for pH of 6.5 and 9.0 Standard Units (S.U.), respectively. Compliance with these limits is measured as daily grab samples. Levels of pH in the Outfall 001 final effluent from June 2013 through October 2017 ranged from 6.3 to 8.1 S.U., averaging 7.2 S.U. There were three days when the daily minimum pH limit of 6.5 S.U. was exceeded. Based on this information, we recommend the draft permit retain the current final effluent limits and monitoring requirements for pH.

4) The current NPDES permit includes a monthly average total phosphorus concentration limit of 0.5 milligram per liter (mg/l) and a loading limit of 13 pounds per day (lbs/day) from May 1 through March 31 and a monthly average concentration limit of 0.65 mg/l and a loading limit of 17 lbs/day from April 1 through April 30. Compliance is determined as 24-hour composite samples collected five times per week. Concentrations and loading of total phosphorus in the Outfall 001 final effluent from June 2013 through October 2017 ranged from 0.03 to 4.2 mg/l, averaging 0.2 mg/l as a concentration and 0.48 to 107 lbs/day, averaging 4.9 lbs/day as a loading. The monthly average concentration was exceeded one time during this time period in October 2014. The load limit was never exceeded during this time period. To protect the Clam River and downstream surface waters from excessive nutrients, we recommend the draft NPDES permit retain the current limitations for total phosphorus.
   a. The load limit of 13 lbs/day is based on the current design flow of 3.2 MGD, the total phosphorus concentration limit of 0.5 mg/l, and 8.34 (conversion factor).
   b. The load limit of 17 lbs/day is based on the current design flow of 3.2 MGD, the total phosphorus concentration limit of 0.65 mg/l, and 8.34 (conversion factor).
   c. Historically, the Clam River was known to have problems with excessive plant growth due to excessive nutrients from point source discharges. The limitations for total phosphorus have been in place for several permit cycles.

5) The current NPDES permit does not include any monitoring requirements or final effluent limitations for total residual chlorine. In fact, the current NPDES permit indicates ultraviolet disinfection is used. The flow diagram and the brochure describing the history and treatment provided in the NPDES permit application indicates chlorine is used for disinfection. MDEQ District staff confirmed that ultraviolet disinfection is used and that no chlorine is used and instructed the Cadillac WWTP to update their flow diagram and facility brochure. Based on this information, we have no recommendations for total residual chlorine as it is understood ultraviolet disinfection is used. However, if chlorine is used, we recommend the draft NPDES permit to include a daily maximum WQBEL for total residual chlorine of 38 micrograms per liter (µg/l). If this limit is included, compliance with this limit shall be daily with samples collected as grab samples and analyzed using an approved USEPA method.
   a. A telephone call with Donal Brady (MDEQ Cadillac District Office) on 1/5/2018 indicated ultraviolet disinfection is used and not chlorine for disinfection. He sent an email to the Cadillac WWTP instructing the facility to update their flow diagram and brochure as it appears to be outdated (See email dated 1/5/2018 below).

6) The current NPDES permit includes a monthly average WQBEL for available cyanide of 5.9 µg/l (0.16 lbs/day) with compliance determined as monthly grab samples. Concentrations of available cyanide in the Outfall 001 final effluent from June 2013 through October 2017 ranged from not detected (<2.0 ug/l) to 5.0 ug/l. These data indicate there is not a reasonable potential for available cyanide to be discharged at concentrations exceeding Michigan WQS. We recommend the draft NPDES permit remove the final effluent limitations and monitoring requirements for available cyanide.

7) The current NPDES permit includes a quarterly monitoring requirement for total copper. Outfall 001 final effluent concentrations for total copper collected from June 2013 through October 2017 ranged from not detected (<50 ug/l) to 86 ug/l. It should be noted the permittee reports a quantification level of 50 ug/l for the analysis of total copper. According to information provided during the previous WQBEL review, the Cadillac WWTP has the capability to analyze copper using its Atomic Absorption machine (flame method) which can reach a detection limit of 50 ug/l. This analytical method and quantification level is not sufficiently sensitive for the characterization of total copper in the Outfall 001 final effluent as the WQBELs for total copper are less than 50 ug/l. Cadillac District staff confirmed the facility is reporting a quantification level of 50 ug/l and has instructed the facility to achieve a much lower quantification level. Since the final effluent data reported as not detected using a quantification level of 50 ug/l are not sufficiently sensitive, these data were not used in the reasonable potential analysis. Based on this information, there is a reasonable potential for total copper to be discharged at concentrations exceeding Michigan WQS. We recommend the draft NPDES permit include a monthly average and daily maximum WQBELs of 11 ug/l (0.3 lbs/day) and 31 ug/l (0.82 lbs/day), respectively. Compliance with these limitations shall be weekly with samples collected as 24-hour composite samples and analyzed using an approved USEPA method found in 40 CFR Part 136 capable of achieving a quantification level of 1.0 ug/l. Appropriate language shall also be included in the draft NPDES permit to allow a monitoring frequency reduction from weekly to no less than quarterly after one year of testing has been completed.
   a. Weekly monitoring is being recommend as several of the results exceeded both chronic and acute levels.

8) Final effluent data for total nickel ranged from not detected (<5.0 ug/l) to 85 ug/l. These data indicate there is a reasonable potential for total nickel to be discharged at concentrations exceeding Michigan WQS. We recommend the draft NPDES permit include a monthly average WQBEL for total nickel of 51 ug/l (1.4 lbs/day). Compliance with this limitation shall be weekly with samples collected as 24-hour composite samples and analyzed using an approved USEPA method capable of achieving a quantification level of 5.0 ug/l. Appropriate language shall also be included in the draft NPDES permit to allow a monitoring frequency reduction from weekly to no less than quarterly after one year of testing has been completed.
   a. Weekly monitoring is being recommended to characterize the final effluent for total nickel over a one year period and to determine compliance with the WQBEL.

9) Final effluent data for total selenium ranged from not detected (<1.0 ug/l) to 6.6 ug/l. These data indicate there is a reasonable potential for total selenium to be discharged at concentrations exceeding Michigan WQS. We recommend the draft NPDES permit include a monthly average WQBEL for total selenium of 5.7 ug/l (0.15 lbs/day). Compliance with this limitation shall be weekly with samples collected as 24-hour composite samples and analyzed using an approved USEPA method capable of achieving a quantification level of 1.0 ug/l. Appropriate language shall also be included in the draft NPDES permit to allow a monitoring frequency reduction from weekly to no less than quarterly after one year of testing has been completed.
   a. Weekly monitoring is being recommended to characterize the final effluent for total selenium over a one year period and to determine compliance with the WQBEL.

10) Final effluent data for bis(2-ethylhexyl)phthalate ranged from not detected (<5.0 ug/l) to 19 ug/l. These data indicate there is a reasonable potential for bis(2-ethylhexyl)phthalate to be discharged at concentrations exceeding Michigan WQS. We recommend the draft NPDES permit include a monthly average WQBEL for bis(2-ethylhexyl)phthalate of 31 ug/l (0.82 lbs/day). Compliance with this limitation shall be weekly with samples collected as 24-hour composite samples and analyzed using an approved USEPA method capable of achieving a quantification level of 5.0 ug/l. Appropriate language shall also be included in the draft NPDES permit to allow a monitoring frequency reduction from weekly to no less than quarterly after one year of testing has been completed.
   a. Weekly monitoring is being recommended to characterize the final effluent for bis(2-ethylhexyl)phthalate over a one year period and to determine compliance with the WQBEL.

11) The current NPDES permit includes a 12-month rolling average level currently achievable (LCA) final effluent limit for total mercury of 2.0 nanograms per liter (ng/l) (0.000053 lbs/day) with compliance samples collected as quarterly grab samples. Part I.A.3 of the current NPDES permit also describes the requirements for the Pollutant Minimization Program (PMP) for total mercury. Concentrations of total mercury in the Outfall 001 final effluent from June 2013 through October 2017 ranged from not detected (<0.5 ng/l) to 1.1 ng/l. Based on the final effluent data for total mercury, there is a reasonable potential for total mercury to be discharged from Outfall 001 at levels exceeding Michigan WQS. We recommend the draft permit retain the current LCA final effluent limit for total mercury of 2.0 ng/l (0.000053 lbs/day). Compliance with this limit should continue on a quarterly basis with samples collected as grab samples using USEPA method 1669 and analyzed using USEPA method 1631 with a quantification level of 0.5 ng/l. In addition, we recommend the PMP requirements be retained in the draft permit.
   a. Review of total mercury final effluent data and reasonable potential analysis was consistent with Water Resources Division, Policy and Procedure, WRD-004, Calculation of Level Currently Achievable (LCA) for Mercury in Proposed National Pollutant Discharge Elimination System (NPDES) Permits.
   b. Quarterly monitoring is recommended to continue to ensure concentrations of total mercury remain at or below the WQS of 1.3 ng/l.

12) The current NPDES permit includes a chronic Whole Effluent Toxicity (WET) final effluent limitation of 1.1 chronic toxic units (TUc) and monitoring for acute WET. Compliance with this limitation is conducted as quarterly 24-hour composite samples and WET testing using *Ceriodaphnia dubia* (*C. dubia*) as the test species using chronic toxicity testing methods. Chronic and acute WET testing using fathead minnow as the test species is required in Part I.A.2, Additional Monitoring Requirements. From June 2013 through October 2017, the Outfall 001 final effluent was never acute or chronically toxic to either test species as all test results report 0.0 acute toxic units (TUa) and 0.0 TUc. Based on this information, there is not a reasonable potential for the Outfall 001 final effluent to be acutely or chronically toxic. We recommend the draft NPDES permit remove the chronic toxicity limit and reporting of the acute toxicity results. We do recommend that acute and chronic WET testing be included in the conditions in Part I.A.2, Additional Monitoring Requirements with testing conducted using both fathead minnow and *C. dubia* as the test species for a total of eight tests (four tests on each species) using chronic toxicity testing methods and samples collected as 24-hour composite samples.

13) The analytical data provided in the permit application for the pollutants listed in the table below were all analyzed using an analytical method and quantification level that was not sufficiently sensitive to determine if these pollutants have the reasonable potential to be discharged at levels below Michigan WQS. This occurred during three of the four final effluent scans collected as part of Part I.A.2, Additional Monitoring Requirements. Based on this information, we recommend the draft permit include a quarterly monitoring requirement for these pollutants. Samples for these pollutants shall be collected as grab samples and analyzed using the following USEPA methods and quantification levels in the table below. Appropriate language should also be included in the draft permit to allow the permittee to request a monitoring frequency after 12-months to reduce the monitoring frequency to no less than annually. Alternate methods may be requested as long as they are USEPA approved methods that can meet the quantification level specified.

Parameter	CAS Number	EPA Method	Quantification Level (µg/L)
3,3-Dichlorobenzidine	91941	605	1.5
Benzidine	92875	605	0.1
Hexachlorobenzene	118741	612	0.01
Hexachlorobutadiene	87683	612	0.01
Hexachlorocyclopentadiene	77474	612	0.01
Acrylonitrile	107131	EPA Approved	1.0
4-Chloro-3-methylphenol	59507	EPA Approved	5.0
Fluoranthene	206440	EPA Approved	1.0
Pentachlorophenol	87865	EPA Approved	1.8
Phenanthrene	85018	EPA Approved	1.0
2,4,6-Trichlorophenol	88062	EPA Approved	5.0

14) The recommendations contained herein are based on the Part 4 Water Quality Standards and Part 8 Water Quality-Based Effluent Limit Development for Toxic Substances. Treatment methods or economic concerns have not been addressed. We understand that other considerations may be used in deciding permit limits and requirements.

15) All other data reviewed indicated there was not a reasonable potential for these parameters to be discharged at concentrations exceeding Michigan WQS. These data were either provided with the NPDES permit application, in the Permits Section toxics facility file, or in MiWaters. Parameters with results reported as all Not Detected were also reviewed. Based on this information, we have no recommendations for these parameters at this time. Refer to the PEL sheet and permit application for a list of other parameters reviewed.

NPDES PERMIT REVIEW SUMMARY

Facility:	Cadillac WWTP
Permit Number:	MI0020257
Reviewer:	Glen Schmitt, Lakes Michigan and Superior Permits Unit, Permits Section, WRD, MDEQ
Date:	1/30/2018

	Authorized Flow		**Disch.**			**FLOWS (cfs)**				
Outf.	**mgd**	**cfs**	**Type ***	**Receiving Water**	**Lake?**	**95% Exc.**	**H. Mean**	**90Q10**	**pH**	**Hardness**
001	3.2	4.95		Clam River	☐	2.9	14.0	3.6	7	75

* P = Process; N = Noncontact Cooling; C = Contact Cooling; S = Sanitary; ST = Storm Water; GW = Groundwater Purge

SOURCES REVIEWED

X **Permit Application** — Type of issuance: REISSUANCE

X **Facility/VGW File** — Low Flow File No. 8970

X **MiWaters Documents & Files** — Hardness value collected from Clam River during 10/14/2014 Point Source Monitoring Survey

X **Current Permit** — Parameters Limited:

Parameter	Mo Avg.	Load	D Max

Special Conditions:

WTA — WTA/NMS GUIDANCE

X **Biosurvey Reports** — Report #:

X **WET Tests** — Date:

X **MiSWIMS/STORET**

MiSWIM Surface Water Information Management System

STORET

REASONABLE POTENTIAL: CHEMICAL SPECIFIC (p. 1 of 2)

*concentration values in ug/L except as noted

Facility:	Cadillac WWTP	Hardness:	75 mg/L CaCO3
Outfall:	001	pH:	7
Date:	1/30/2018	Drinking Water:	n

Parameter	CAS #	Water Quality Values										Diss. Met.	Background
		FCV	vd	HNV	vd	HCV	vd	WV	vd	FAV	vd	Translator	Conc.
Cyanide, free	57125	5.2	1199707	48000	1199707	NA	0	NA	0	44	1199707	1	0
Copper	7440508	7.00393125	1199707	38000	1200512	NA	0	NA	0	20.49667108	1199707	1.5	4
Mercury @	7439976	0.77	1199707	0.0018	1199707	NA	0	0.0013	1199707	2.8	1199707	1	**
Barium	7440393	322.4532343	2200905	160000	1199705	NA	0	NA	0	1840.068569	2200905	1	0
Boron	7440428	7200	1201511	330000	1201511	NA	0	NA	0	69000	1201511	1	0
Cadmium	7440439	1.809112735	1199707	130	1199706	NA	0	NA	0	6.243962757	1199707	2.1	0
Chromium	7440473	58.55695529	1199707	9400	1199706	NA	0	NA	0	900.3259642	1199707	1.5	1.6
Lead	7439921	16.50732835	2201011	190	1200709	NA	0	NA	0	316.5089773	1201011	4.5	2.5
Molybdenum	7439987	3200	2200604	10000	1200605	NA	0	NA	0	58000	2200604	1	0
Nickel	7440020	40.77179058	1199707	210000	1199706	NA	0	NA	0	734.1696421	1199707	1.1	3
Selenium	7782492	5	1199707	2700	1199704	NA	0	NA	0	120	1199808	1	0
Silver	7440224	0.06	1199710	11000	1199705	NA	0	NA	0	1.1	1199710	20	0
Zinc	7440666	92.58329763	1199707	16000	1200510	NA	0	NA	0	183.6642294	1199707	2.1	13
Bis(2-ethylhexyl)phthalate #	117817	ID*	199809	160	1199711	18	1201402	NA	0	285	2199809	1	0
#N/A	1	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A

= Carcinogen

** = Background exceeds standards.

REASONABLE POTENTIAL: CHEMICAL SPECIFIC (p. 2 of 2)

**concentration values in ug/L except as noted; loads in lb/d*

Facility:	Cadillac WWTP	Disch. Rate:	4.95 cfs	95% Ex. Flow:	2.9 cfs	
Outfall:	001			H. Mean Flow:	14 cfs	
Date:	1/30/2018			90Q10 Flow:	3.6 cfs	

	Conc.	Load (lbs/day)
Hg Loading Calculated based on LCA (ng/L) of :		0
TP Loading based on concentration (mg/L) of :		0

Parameter	Monthly Average PEL								Daily Max PEL		PEQ		DECISION	
	FCV	load	HNV	load	HCV	load	WV	load	conc	load	Avg	Max	Avg	Max
Cyanide, free	5.96106875	0.159089	81915	2186.148	#VALUE!	#VALUE!	#VALUE!	#VALUE!	44	1.174272	5.5	5.5	No Reasonable Potential. Remove Limit and monitoring requirements.	
Copper	11.45809587	0.305794	64846.5488	1730.625	#VALUE!	#VALUE!	#VALUE!	#VALUE!	30.74501	0.820523	197.8	197.8	Reasonable potential. Recommend a Mon. Avg. and Daily Max. WQBEL of 11 ug/l (0.3 lbs/day) and 31 ug/l (0.82 lb/day), weekly monitoring.	
Mercury @	0.77	0.02055	0.0018	4.8E-05	NA	#VALUE!	0.0013	3.4694E-05	2.8	0.074726	0.00154	0.00154	Reasonable Potential. Recommend retain current LCA and monitoring requirements and PMP.	
Barium	369.6472882	9.865147	273050	7287.158	#VALUE!	#VALUE!	#VALUE!	#VALUE!	1840.069	49.10775	75.4	75.4	No Reasonable Potential. No recommendations.	
Boron	8253.7875	220.2771	563165.625	15029.76	#VALUE!	#VALUE!	#VALUE!	#VALUE!	69000	1841.472	1378	1378	No Reasonable Potential. No recommendations.	
Cadmium	4.355176023	0.116231	221.853125	5.920816	#VALUE!	#VALUE!	#VALUE!	#VALUE!	13.11232	0.349942	1.173	1.173	No Reasonable Potential. No recommendations.	
Chromium	100.456797	2.680991	16040.557	428.0904	#VALUE!	#VALUE!	#VALUE!	#VALUE!	1350.489	36.04185	13.552	13.552	No Reasonable Potential. No recommendations.	
									PEQ >= +10 Samples		2.149806	4.618595		
Lead	84.78908932	2.262851	322.480469	8.606359	#VALUE!	#VALUE!	#VALUE!	#VALUE!	1424.29	38.01146	4.505	4.505	No Reasonable Potential. No recommendations.	
Molybdenum	3668.35	97.90092	17065.625	455.4474	#VALUE!	#VALUE!	#VALUE!	#VALUE!	58000	1547.904	14.598	14.598	No Reasonable Potential. No recommendations.	
									PEQ >= +10 Samples		7.711422	12.76473		
Nickel	50.97395868	1.360393	358376.005	9564.339	#VALUE!	#VALUE!	#VALUE!	#VALUE!	807.5866	21.55287	102	102	Reasonable Potential. Recommend a Mon. Avg. WQBEL of 51 ug/l (1.4 lbs/day), weekly monitoring.	
Selenium	5.731796875	0.15297	4607.71875	122.9708	#VALUE!	#VALUE!	#VALUE!	#VALUE!	120	3.20256	11.88	11.88	Reasonable Potential. Recommend a Mon. Avg. WQBEL of 5.7 ug/l (0.15 lbs/day), weekly monitoring.	
Silver	1.37563125	0.036713	18772.1875	500.9921	#VALUE!	#VALUE!	#VALUE!	#VALUE!	22	0.587136	2.528	2.528	No Reasonable Potential. No recommendations.	
									PEQ >= +10 Samples		0.719367	1.555352		
Zinc	220.9781637	5.897465	27295.8147	728.4707	#VALUE!	#VALUE!	#VALUE!	#VALUE!	385.6949	10.29343	115.5	115.5	No Reasonable Potential. No recommendations.	
Bis(2-ethylhexyl)phthalate #	#VALUE!	#VALUE!	273.05	7.287158	30.71813	0.819805	#VALUE!	#VALUE!	285	7.60608	49.4	49.4	Reasonable Potential. Recommend a Mon. Avg. WQBEL of 31 ug/l (0.82 lbs/day), weekly monitoring.	
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		

** SILVER - Appropriate Dissolved Metal Translator must be entered in PEL 1*

PEQ -- Effluent Data Summary
(all values in ug/L)

DATA ENTRY

Parameter:	Cyanide, free	Source	Copper	Source	Mercury @	Source	Barium	Source	Boron	Source	Cadmium	Source	Chromium	Source	Lead	Source	Molybdenum	Source
No. Nondetect	**46**	**DMRs**	**0**	**DMRs**	**13**	**DMRs**	**0**	**App.**	**0**	**App.**	**4**	**App./MiWaters**	**3**	**IPP Data/App.**	**4**	**IPP/App.**	**0**	**IPP data**
Data	5	12/2/2015	86	1/3/2014	0.0011	10/6/2015	22	5/2/2016	370	5/2/2016	0.25	Mar-15	1.2	Jan-15	2.04	May-15	5.44	Jan-15
	2	10/10/2016	54	6/22/2014	0.0011	11/3/2015	20	8/1/2016	320	8/1/2016	0.31	Apr-15	1.61	Feb-15	2.1	Sep-15	6.09	Feb-15
			7.3	10/15/2014 CSI	0.0006	5/11/2016	29	10/6/2016	530	10/6/2016	0.35	May-15	1.01	Mar-15	1.75	Nov-15	4.22	Mar-15
			56	5/22/2015	0.000525	2/6/2017	20	2/2/2017	480	2/2/2017	0.21	Jun-15	8.47	Jun-15	2.65	Dec-15	4.74	Apr-15
			76	6/13/2015							0.2	Oct-15	1.19	Jul-15	2.14	May-16	8.11	May-15
											0.69	Nov-15	1.28	Sep-15	2.03	Dec-16	6.99	Jun-15
											0.21	May-16	1.49	Dec-15			4.22	Jul-15
													1.05	Jan-16			5.14	Aug-15
		Not Detected				Not Detected							2.12	Feb-16			5.55	Sep-15
		8/1/2013		There were several		11/4/2013							2.71	Mar-16			6.69	Oct-15
		11/4/2013		final effluent results		2/4/2014							1.29	Apr-16			3.38	Nov-15
		2/3/2014		for total copper		5/6/2014						Not Detected	1.6	May-16		Not Detected	5.64	Dec-15
		4/7/2014		reported as <50 ug/l.		8/6/2014						5/2/2016	1.73	Jun-16		5/2/2016	3.8	Jan-16
		5/5/2014		These data were not		11/12/2014						8/1/2016	1.8	Jul-16		8/1/2016	3.08	Feb-16
		6/3/2014		used as the method		2/3/2015						2/2/2017	1.48	Aug-16		2/2/2017	9.3	Mar-16
		7/1/2014		and QL used are		5/8/2015						10/6/2016	1.68	Sep-16		10/6/2016	5.61	Apr-16
		8/6/2014		not sufficiently		8/5/2015							1.29	Oct-16			9.62	May-16
		9/3/2014		sensitive.		2/8/2016							1.37	Dec-16			16.93	Jun-16
		10/7/2014				8/2/2016						QL <0.2 ug/l	1	2/2/2017		QL <1.0 ug/l	9.52	Jul-16
		10/14/2014				11/7/2016											13.1	Aug-16
		11/3/2014				5/22/2017											7.96	Sep-16
		12/1/2014				8/10/2017						There was					4.57	Oct-16
		1/6/2015										several cadmium					5.15	Nov-16
		2/3/2015										data points					6.12	Dec-16
		3/2/2015										with a result		Not Detected		There were a		
		4/1/2015										of "0". These		5/1/2016		few results		
		5/5/2015										data were not		8/1/2016		reported as "0"		
		6/8/2015										used since the		10/6/2016		that were not		
		7/1/2015				QL <0.5 ng/l						QL could				used since the		
		8/1/2015										not be verified.				QL could not be		
		9/14/2015														verified.		
		10/3/2015												QL <5.0 ug/l				
		11/3/2015																
		1/4/2016																
		2/1/2016												There were a				
		3/7/2016												few results				
		4/12/2016												reported as "0"				
		5/2/2016												that were not				
		6/8/2016												used since the				
		7/6/2016												QL could not be				
		8/1/2016												verified.				
		9/7/2016																
		11/1/2016																
		12/6/2016																
		1/4/2017																
		2/2/2017																
		3/13/2017																
		4/13/2017																
		5/16/2017																
		5/16/2017																
		6/15/2017																
		7/4/2017																
		8/10/2017																
		9/25/2017																
		10/24/2017																
		QL <2.0 ug/l																

PEQ -- Effluent Data Summary
(all values in ug/L)

Parameter:	Nickel	Source	Selenium	Source	Silver	Source	Zinc	Source	Bis(2-ethylhexyl)phthalate #	Source	#N/A	Source	#N/A	Source	#N/A	Source	#N/A
No. Nondetect	**24**	**Ipp Data/App.**	**6**	**DMRs**	**4**	**App./MiWaters**	**0**	**DMRs/App.**	**3**	**App.**	**0**		**0**		**0**		**0**
Data	5.1	5/2/2016	1.9	5/2/2016	0.2	Jan-15	51.2	May-15	19	5/2/2016							
	85	May-16	6.6	8/1/2016	0.55	Feb-15	55	Jun-15									
	5.5	8/1/2016	4	10/6/2016	1.55	Mar-15	40	5/2/2016									
	5	2/2/2017			1.58	Apr-15	35	8/1/2016									
					0.77	Jun-15	29	10/6/2016									
					0.66	Jul-15	22	2/2/2017									
					0.34	Aug-15				Not Detected							
					1.18	Sep-15				8/1/2016 (<10 ug/l)							
					0.26	Jun-16				10/6/2016 (<10 ug/l)							
				Not Detected	0.02	Jul-16				2/2/2017 (<5.0 ug/l)							
				8/7/2013	0.37	Oct-16		There was									
		Not Detected		8/10/2013	1.14	Dec-16		several zinc									
		10/6/2016 (<5.0 ug/l)		8/14/2013				data points									
		Jan-15		8/22/2013				with a result									
		Feb-15		8/27/2013				of "0". These									
		Mar-15		2/2/2017				data were not									
		Apr-15						used since the									
		May-15						QL could									
		Jun-15						not be verified.									
		Jul-15				Not Detected											
		Aug-15		QL <1.0 to <2.0 ug/l		5/1/2016											
		Sep-15				8/1/2016											
		Oct-15				10/6/2016											
		Nov-15				2/2/2017											
		Dec-15															
		Jan-16															
		Feb-16															
		Mar-16				QL <0.2 to <0.5 ug/l											
		Apr-16															
		Jun-16															
		Jul-16				There was											
		Aug-16				several silver											
		Sep-16				data points											
		Oct-16				with a result											
		Nov-16				of "0". These											
		Dec-16				data were not											
						used since the											
						QL could											
		Rest of dates used a				not be verified.											
		QL <50 ug/l															

PROJECTED EFFLUENT QUALITY ANALYSIS (<10 Detectable):

* all values in ug/l

	Cyanide, free	Copper	Mercury @	Barium	Boron	Cadmium	Chromium	Lead	Molybdenum	Nickel	Selenium	Silver	Zinc	Bis(2-ethylhexyl)phthalate #	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
Sample Max.	5	86	0.0011	29	530	0.69	8.47	2.65	8.11	85	6.6	1.58	55	19	0	0	0	0	0	0
Total Samples	48	5	17	4	4	11	12	10	9	28	9	13	6	4	0	0	0	0	0	0
Multiplier	1.1	2.3	1.4	2.6	2.6	1.7	1.6	1.7	1.8	1.2	1.8	1.6	2.1	2.6	0	0	0	0	0	0
PEQ:	**5.5**	**197.8**	**0.00154**	**75.4**	**1378**	**1.173**	**13.552**	**4.505**	**14.598**	**102**	**11.88**	**2.528**	**115.5**	**49.4**	**0**	**0**	**0**	**0**	**0**	**0**

PROJECTED EFFLUENT QUALITY ANALYSIS (>=10 Detectable):

PARAMETER:	Cyanide, free		Copper		Mercury @		Barium		Boron		Cadmium	
TOTAL NO. VALUES:	**48**		**5**		**17**		**4**		**4**		**11**	
DETECTED:	**2**		**5**		**4**		**4**		**4**		**7**	
NON-DETECTED:	**46**		**0**		**13**		**0**		**0**		**4**	
d (% data<detect)	0.958333333		0		0.764706		0		0		0.3636364	
m (mean of data >detect)	3.5		55.86		0.000831		22.75		425		0.3171429	
std dev	2.121320344		30.31333		0.000312		4.272002		96.781541		0.17385	
n	1	30	1	30	1	4	1	30	1	30	1	30
d^n	0.958333333	0.278932	0	0	0.764706	0.341962	0	0	0	0	0.3636364	6.607E-14
p	-0.2	0.930658	0.95	0.95	0.7875	0.924017	0.95	0.95	0.95	0.95	0.9214286	0.95
std normal dist fn (Z)	#NUM!	2.310286	2.447747	2.447747	1.760008	2.270348	2.447747	2.447747	2.4477468	2.4477468	2.2555475	2.4477468
Z_p	#NUM!	1.481002	1.645211	1.645211	0.797575	1.432884	1.645211	1.645211	1.6452114	1.6452114	1.4150024	1.6452114
1+(s/m)^2	1.367346939	1.367347	1.294487	1.294487	1.140728	1.140728	1.035261	1.035261	1.051857	1.051857	1.3004964	1.3004964
(sigma_d)^2	0.312872321	0.312872	0.258114	0.258114	0.131666	0.131666	0.034654	0.034654	0.0505572	0.0505572	0.2627461	0.2627461
mu_d	1.096326808	1.096327	3.893792	3.893792	-7.158413	-7.158413	3.107238	3.107238	6.0268106	6.0268106	-1.279776	-1.279776
(sigma_dn)^2	0.312872321	0.395949	0.258114	0.009768	0.131666	0.255484	0.034654	0.001175	0.0505572	0.0017271	0.2627461	0.0341965
mu_dn	1.096326808	-1.796244	3.893792	4.017964	-7.158413	-8.248748	3.107238	3.123978	6.0268106	6.0512256	-1.279776	-1.617486
P_95 exponent	#NUM!	-0.864331	4.72964	4.180569	-6.869006	-7.524491	3.413504	3.180365	6.396735	6.1195975	-0.554463	-1.313249
P_95 (PEQ)	**#NUM!** Max	**0.421333** Average	**113.2548** Max	**65.40304** Average	**0.00104** Max	**0.00054** Average	**30.37148** Max	**24.05554** Average	**599.88324** Max	**454.68163** Average	**0.5743805** Max	**0.2689448** Average

NOTES: For purposes of this summary, ^ represents an exponent or superscript while _ represents a subscript.
Variables are defined in Part 12, Rule 1225 (3) (a).

Chromium		Lead		Molybdenum		Nickel		Selenium		Silver		Zinc	
22		**10**		**24**		**28**		**9**		**16**		**6**	
19		**6**		**24**		**4**		**3**		**12**		**6**	
3		**4**		**0**		**24**		**6**		**4**		**0**	
0.1363636		0.4		0		0.85714286		0.66666667		0.25		0	
1.8615789		2.1183333		6.70708333		25.15		4.16666667		0.71833333		38.7	
1.6528671		0.2943071		3.20660254		39.9005848		2.35442845		0.53004002		12.7302789	
1	30	1	30	1	30	1	30	1	30	1	30	1	30
0.1363636	1.099E-26	0.4	1.153E-12	0	0	0.85714286	0.00980836	0.66666667	5.2151E-06	0.25	8.67362E-19	0	0
0.9421053	0.95	0.9166667	0.95	0.95	0.95	0.65	0.94950472	0.85	0.949999739	0.93333333	0.95	0.95	0.95
2.3871023	2.4477468	2.2293078	2.4477468	2.44774683	2.44774683	1.44901492	2.44371664	1.94788089	2.4477447	2.32725168	2.447746831	2.44774683	2.44774683
1.5730254	1.6452114	1.3832326	1.6452114	1.64521144	1.64521144	0.38487708	1.64042643	1.03643149	1.645208911	1.50138519	1.64521144	1.64521144	1.64521144
1.7883389	1.7883389	1.0193025	1.0193025	1.22857191	1.22857191	3.5169961	3.5169961	1.319296	1.319296	1.54445913	1.544459132	1.10820664	1.10820664
0.5812872	0.5812872	0.0191185	0.0191185	0.20585245	0.20585245	1.25760725	1.25760725	0.27709826	0.277098261	0.43467377	0.434673772	0.10274307	0.10274307
0.3307814	0.3307814	0.7410703	0.7410703	1.80023796	1.80023796	2.59605427	2.59605427	1.28856723	1.288567225	-0.54815845	-0.548158451	3.60446807	3.60446807
0.5812872	0.0350681	0.0191185	0.0230274	0.20585245	0.00759019	1.25760725	0.57084882	0.27709826	0.094028029	0.43467377	0.034700218	0.10274307	0.0036004
0.3307814	0.4572875	0.7410703	0.2282903	1.80023796	1.89936909	2.59605427	1.00338012	1.28856723	0.281495268	-0.54815845	-0.635853746	3.60446807	3.6540394
1.5300905	0.7653778	0.9323297	0.4779476	2.54668627	2.04270266	3.02766733	2.24279672	1.83414604	0.785981964	0.44170171	-0.329383851	4.13181689	3.75275755
4.6185948	**2.1498064**	**2.5404206**	**1.6127609**	**12.7647347**	**7.71142238**	**20.649009**	**9.41963857**	**6.25978625**	**2.194560864**	**1.55535172**	**0.719366834**	**62.2909963**	**42.638498**
Max	Average	Max	Average	Max	Average	Max	Average	Max	Average	Max	Average	Max	Average

Bis(2-ethylhexyl)phthalate #		**#N/A**		**#N/A**		**#N/A**		**#N/A**		**#N/A**		**#N/A**
4		**0**		**0**		**0**		**0**		**0**		**0**
1		**0**		**0**		**0**		**0**		**0**		**0**
3		**0**		**0**		**0**		**0**		**0**		**0**
0.75		#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!
19		#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!
#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!
1	30	1	30	1	30	1	30	1	30	1	30	1
0.75	0.00017858	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
0.8	0.94999107	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
1.794122578	2.44767387	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
0.841456717	1.64512482	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
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#DIV/0!	**#DIV/0!**	**#DIV/0!**	**#DIV/0!**	**#DIV/0!**	**#DIV/0!**	**#DIV/0!**	**#DIV/0!**	**#DIV/0!**	**#DIV/0!**	**#DIV/0!**	**#DIV/0!**	**#DIV/0!**
Max	Average	Max	Average	Max	Average	Max	Average	Max	Average	Max	Average	Max

REASONABLE POTENTIAL ANALYSIS: WET

Facility: Cadillac WWTP
Outfall: 001

			PEL		PEQ*	
Qe	Qr	RWC	TU_C	TU_A	TU_C	TU_A
4.953560372	0.725	87.23	1.1463594	1	0.00	0.00

* PEQ (enter data for most sensitive species only):

Chronic

Total n =	25	Quant n =	0	CV =	0.6
TUc effluent =	0	MF =	1.4	PEQ TUc =	0.00

Acute

Total n =	25	Quant n =	0	CV =	0.6
TU_A effluent =	0	MF =	1.4	PEQ TU_A =	0.00

Decision (WET limits/conditions):

WQBEL				Monitoring	
Monthly Avg	Daily Max	Species	TI/RE	Short term	Long term

Basis of decision:

No RP as all tests for both species demonstrated no acute or chronic toxicity. Recommend acute and chronic WET testing be required under the Additional Monitoring Requirements with four tests conducted on both Fathead minnow and Ceriodaphnia dubia for a total of eight tests using chronic toxicity testing methods.

WET Data Summary

Species 1: Ceriodaphnia dubia

		TU		
Lab	Test Date	Acute	Chronic	Reference
	6/6/2013	0.0	0.0	MiWaters
	7/9/2013	0.0	0.0	MiWaters
	8/8/2013	0.0	0.0	MiWaters
	9/10/2013	0.0	0.0	MiWaters
	10/3/2013	0.0	0.0	MiWaters
	11/14/2013	0.0	0.0	MiWaters
	12/5/2013	0.0	0.0	MiWaters
	1/7/2014	0.0	0.0	MiWaters
	2/13/2014	0.0	0.0	MiWaters
	3/6/2014	0.0	0.0	MiWaters
	4/3/2014	0.0	0.0	MiWaters
	8/7/2014	0.0	0.0	MiWaters
	11/13/2014	0.0	0.0	MiWaters
Paragon	2/5-2/12/2015	0.0	0.0	MiWaters
Paragon	5/7-5/14/2015	0.0	0.0	MiWaters
Paragon	8/6-8/12/2015	0.0	0.0	MiWaters
Paragon	11/5-11/12/2015	0.0	0.0	MiWaters
Paragon	2/4-2/11/2016	0.0	0.0	MiWaters
Paragon	5/5-5/12/2016	0.0	0.0	MiWaters
Paragon	8/4-8/10/2016	0.0	0.0	MiWaters
Paragon	10/14-10/20/2016	0.0	0.0	MiWaters
Paragon	11/3-11/10/2016	0.0	0.0	MiWaters
ERM	2/2-2/8/2017	0.0	0.0	MiWaters
	5/2/2017	0.0	0.0	MiWaters
	8/8/2017	0.0	0.0	MiWaters

Species 2: Fathead minnow

		TU		
Lab	Test Date	Acute	Chronic	Reference
	5/6-5/7/2015	0.0	0.0	MiWaters
Paragon	5/5-5/12/2016	0.0	0.0	MiWaters
Paragon	8/4-8/11/2016	0.0	0.0	MiWaters
Paragon	10/14-10/21/2016	0.0	0.0	MiWaters
ERM	2/2-2/9/2017	0.0	0.0	MiWaters

NPDES PERMIT REVIEW SUMMARY

Facility:	Cadillac WWTP
Permit Number:	MI0020257
Reviewer:	Glen Schmitt, Lakes Michigan and Superior Permits Unit, Permits Section, WRD, MDEQ
Date:	1/30/2018

	Authorized Flow		**Disch.**			**FLOWS (cfs)**				
Outf.	**mgd**	**cfs**	**Type ***	**Receiving Water**	**Lake?**	**95% Exc.**	**H. Mean**	**90Q10**	**pH**	**Hardness**
001	3.2	4.95		Clam River	☐	2.9	14.0	3.6	7	75

* P = Process; N = Noncontact Cooling; C = Contact Cooling; S = Sanitary; ST = Storm Water; GW = Groundwater Purge

SOURCES REVIEWED

X **Permit Application** — Type of issuance: REISSUANCE

X **Facility/VGW File** — Low Flow File No. 8970

X **MiWaters Documents & Files** — Hardness value collected from Clam River during 10/14/2014 Point Source Monitoring Survey

X **Current Permit** — Parameters Limited:

Parameter	Mo Avg.	Load	D Max

Special Conditions:

WTA — WTA/NMS GUIDANCE

X **Biosurvey Reports** — Report #:

X **WET Tests** — Date:

X **MiSWIMS/STORET**

MiSWIM Surface Water Information Management System

STORET

REASONABLE POTENTIAL: CHEMICAL SPECIFIC (p. 1 of 2)

*concentration values in ug/L except as noted

Facility:	Cadillac WWTP	Hardness:	75 mg/L CaCO3
Outfall:	001	pH:	7
Date:	1/30/2018	Drinking Water:	n

Parameter	CAS #	Water Quality Values										Diss. Met.	Background
		FCV	vd	HNV	vd	HCV	vd	WV	vd	FAV	vd	Translator	Conc.
3,3'-Dichlorobenzidine #	91941	4.5	2199710	950	2199709	0.2	2199709	NA	0	81	2199710	1	0
Benzidine #	92875	2.7	2200902	3700	1201003	0.073	1201003	NA	0	49	2200902	1	0
Hexachlorobenzene # @	118741	ID*	199801	0.046	1199707	0.00045	1199707	0.0003	1199704	1200	**3199801**	1	0
Hexachlorobutadiene # @	87683	1	2199904	0.098	1199704	0.35	2199704	0.053	1199906	15	2199904	1	0
Hexachlorocyclopentadiene	77474	0.07	**3199904**	450	1199904	NA	0	NA	0	0.54	**3199904**	1	0
Nitrobenzene #	98953	230	2201002	990	1201005	180	1201005	NA	0	2100	2201002	1	0
Acrylonitrile #	107131	66	2201010	320	1200708	1.2	1200708	NA	0	1200	2201010	1	0
Acenaphthylene	208968	6.6	3201411	530	**3199803**	NA	0	NA	0	120	3201411	1	0
Azobenzene #	103333	4.1	3201004	ID*	201004	6	2201004	NA	0	74	3201004	1	0
Butyl benzyl phthalate	85687	67	2200207	160	1199902	NA	0	NA	0	630	1200207	1	0
4-Chloro-3-methylphenol	59507	7.4	2200103	39000	1200103	NA	0	NA	0	130	2200103	1	0
Di-n-butyl phthalate	84742	9.7	2199809	690	2199809	NA	0	NA	0	75	2199809	1	0
Fluoranthene	206440	1.6	2201411	18	2199901	NA	0	NA	0	28	1201411	1	0
Hexachloroethane #	67721	13	2200512	7.6	1199707	6.7	1199707	NA	0	210	2200512	1	0
Pentachlorophenol #	87865	6.692583681	1199707	450	1199710	2.8	1199710	NA	0	17.44664176	1199707	1	0
Phenanthrene	85018	1.7	2201411	ID*	199706	NA	0	NA	0	11	2201411	1	0
2,4,6-Trichlorophenol #	88062	5	2200609	14000	1200612	290	1200612	NA	0	79	2200609	1	0
#N/A	1	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A

= Carcinogen

REASONABLE POTENTIAL: CHEMICAL SPECIFIC (p. 2 of 2)

**concentration values in ug/L except as noted; loads in lb/d*

Facility:	Cadillac WWTP	Disch. Rate:	4.95 cfs	95% Ex. Flow:	2.9 cfs		
Outfall:	001			H. Mean Flow:	14 cfs		
Date:	1/30/2018			90Q10 Flow:	3.6 cfs		

	Conc.	Load (lbs/day)
Hg Loading Calculated based on LCA (ng/L) of :		0
TP Loading based on concentration (mg/L) of :		0

Parameter	Monthly Average PEL								Daily Max PEL		PEQ		DECISION	
	FCV	load	HNV	load	HCV	load	WV	load	conc	load	Avg	Max	Avg	Max
3,3'-Dichlorobenzidine #	5.158617188	0.137673	1621.23438	43.2675	0.341313	0.009109	#VALUE!	#VALUE!	81	2.161728	0	0	Not Detected. Sufficiently Sensitive method and QL not used. Recommend Quarterly monitoring.	
Benzidine #	3.095170313	0.082604	6314.28125	168.5155	0.124579	0.003325	#VALUE!	#VALUE!	49	1.307712	0	0	Not Detected. Sufficiently Sensitive method and QL not used. Recommend Quarterly monitoring.	
Hexachlorobenzene # @	#VALUE!	#VALUE!	0.046	0.001228	0.00045	1.2E-05	0.0003	8.0064E-06	1200	32.0256	0	0	Not Detected. Sufficiently Sensitive method and QL not used. Recommend Quarterly monitoring.	
Hexachlorobutadiene # @	1	0.026688	0.098	0.002615	0.35	0.009341	0.053	0.00141446	15	0.40032	0	0	Not Detected. Sufficiently Sensitive method and QL not used. Recommend Quarterly monitoring.	
Hexachlorocyclopentadiene	0.080245156	0.002142	767.953125	20.49513	#VALUE!	#VALUE!	#VALUE!	#VALUE!	0.54	0.014412	0	0	Not Detected. Sufficiently Sensitive method and QL not used. Recommend Quarterly monitoring.	
Nitrobenzene #	263.6626563	7.036629	1689.49688	45.08929	307.1813	8.198053	#VALUE!	#VALUE!	2100	56.0448	0	0	Not Detected. Sufficiently Sensitive method and QL used. No Recommendations.	
Acrylonitrile #	75.65971875	2.019207	546.1	14.57432	2.047875	0.054654	#VALUE!	#VALUE!	1200	32.0256	0	0	Not Detected. Sufficiently Sensitive method and QL not used in 3 of 4 samples. Recommend Quarterly monitoring.	
Acenaphthylene	7.565971875	0.201921	904.478125	24.13871	#VALUE!	#VALUE!	#VALUE!	#VALUE!	120	3.20256	0	0	Not Detected. Sufficiently Sensitive method and QL not used in 3 of 4 samples. Tier III Water Quality Values. No Recommendations.	
Azobenzene #	4.700073438	0.125436	#VALUE!	#VALUE!	10.23938	0.273268	#VALUE!	#VALUE!	74	1.974912	0	0	Not Detected. Sufficiently Sensitive method and QL not used in 3 of 4 samples. Tier III Water Quality Values. No Recommendations.	
Butyl benzyl phthalate	76.80607813	2.049801	273.05	7.287158	#VALUE!	#VALUE!	#VALUE!	#VALUE!	630	16.81344	0	0	Not Detected. Sufficiently Sensitive method and QL used. No Recommendations.	
4-Chloro-3-methylphenol	8.483059375	0.226396	66555.9375	1776.245	#VALUE!	#VALUE!	#VALUE!	#VALUE!	130	3.46944	0	0	Not Detected. Sufficiently Sensitive method and QL not used in 3 of 4 samples. Recommend Quarterly monitoring.	
Di-n-butyl phthalate	11.11968594	0.296762	1177.52813	31.42587	#VALUE!	#VALUE!	#VALUE!	#VALUE!	75	2.0016	0	0	Not Detected. Sufficiently Sensitive method and QL used. No Recommendations.	
Fluoranthene	1.834175	0.04895	30.718125	0.819805	#VALUE!	#VALUE!	#VALUE!	#VALUE!	28	0.747264	0	0	Not Detected. Sufficiently Sensitive method and QL not used in 3 of 4 samples. Recommend Quarterly monitoring.	
Hexachloroethane #	14.90267188	0.397723	12.969875	0.34614	11.43397	0.30515	#VALUE!	#VALUE!	210	5.60448	0	0	Not Detected. Sufficiently Sensitive method and QL used. No Recommendations.	
Pentachlorophenol #	7.672106045	0.204753	767.953125	20.49513	4.778375	0.127525	#VALUE!	#VALUE!	17.44664	0.465616	0	0	Not Detected. Sufficiently Sensitive method and QL not used in 3 of 4 samples. Recommend Quarterly monitoring.	
Phenanthrene	1.948810938	0.05201	#VALUE!	#VALUE!	#VALUE!	#VALUE!	#VALUE!	#VALUE!	11	0.293568	0	0	Not Detected. Sufficiently Sensitive method and QL not used in 3 of 4 samples. Recommend Quarterly monitoring.	
2,4,6-Trichlorophenol #	5.731796875	0.15297	23891.875	637.6264	494.9031	13.20797	#VALUE!	#VALUE!	79	2.108352	0	0	Not Detected. Sufficiently Sensitive method and QL not used in 3 of 4 samples. Recommend Quarterly monitoring.	
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		

** SILVER - Appropriate Dissolved Metal Translator must be entered in PEL 1*

PEQ -- Effluent Data Summary
(all values in ug/L)

DATA ENTRY

Parameter:	3,3'-Dichlorobenzidine #	Source	Benzidine #	Source	Hexachlorobenzene # @	Source	Hexachlorobutadiene # @	Source
No. Nondetect	**4**	**App.**	**4**	**App.**	**4**	**App.**	**4**	**App.**
Data		Not Detected 5/2/2016 (<10 ug/l) 8/1/2016 (<10 ug/l) 10/6/2016 (<10 ug/l) 2/2/2017 (<20 ug/l)		Not Detected 5/2/2016 (<10 ug/l) 8/1/2016 (<10 ug/l) 10/6/2016 (<10 ug/l) 2/2/2017 (<50 ug/l)		Not Detected 5/2/2016 (<10 ug/l) 8/1/2016 (<10 ug/l) 10/6/2016 (<10 ug/l) 2/2/2017 (<5.0 ug/l)		Not Detected 5/2/2016 (<10 ug/l) 8/1/2016 (<10 ug/l) 10/6/2016 (<10 ug/l) 2/2/2017 (<5.0 ug/l)

PEQ -- Effluent Data Summary
(all values in ug/L)

Parameter:	Hexachlorocyclopentadiene	Source	Nitrobenzene #	Source	Acrylonitrile #	Source	Acenaphthylene	Source	Azobenzene #	Source
No. Nondetect	**4**	**App.**	**4**	**App.**	**4**	**App.**	**4**	**App.**	**4**	**App.**
Data		Not Detected 5/2/2016 (<10 ug/l) 8/1/2016 (<10 ug/l) 10/6/2016 (<10 ug/l) 2/2/2017 (<5.0 ug/l)		Not Detected 5/2/2016 (<10 ug/l) 8/1/2016 (<10 ug/l) 10/6/2016 (<10 ug/l) 2/2/2017 (<5.0 ug/l)		Not Detected 5/2/2016 (<5.0 ug/l) 8/1/2016 (<5.0 ug/l) 10/6/2016 (<5.0 ug/l) 2/2/2017 (<1.0 ug/l)		Not Detected 5/2/2016 (<10 ug/l) 8/1/2016 (<10 ug/l) 10/6/2016 (<10 ug/l) 2/2/2017 (<5.0 ug/l)		Not Detected 5/2/2016 (<10 ug/l) 8/1/2016 (<10 ug/l) 10/6/2016 (<10 ug/l) 2/2/2017 (<3.0 ug/l)

PEQ -- Effluent Data Summary
(all values in ug/L)

Parameter:	Butyl benzyl phthalate	Source	4-Chloro-3-methylphenol	Source	Di-n-butyl phthalate	Source	Fluoranthene	Source	Hexachloroethane #	Source
No. Nondetect	**0**		**0**		**0**		**0**		**0**	
Data										
		Not Detected 5/2/2016 (<10 ug/l) 8/1/2016 (<10 ug/l) 10/6/2016 (<10 ug/l) 2/2/2017 (<5.0 ug/l)		Not Detected 5/2/2016 (<10 ug/l) 8/1/2016 (<10 ug/l) 10/6/2016 (<10 ug/l) 2/2/2017 (<5.0 ug/l)		Not Detected 5/2/2016 (<10 ug/l) 8/1/2016 (<10 ug/l) 10/6/2016 (<10 ug/l) 2/2/2017 (<5.0 ug/l)		Not Detected 5/2/2016 (<10 ug/l) 8/1/2016 (<10 ug/l) 10/6/2016 (<10 ug/l) 2/2/2017 (<1.0 ug/l)		Not Detected 5/2/2016 (<10 ug/l) 8/1/2016 (<10 ug/l) 10/6/2016 (<10 ug/l) 2/2/2017 (<5.0 ug/l)

PEQ -- Effluent Data Summary
(all values in ug/L)

Parameter:	Pentachlorophenol #	Source	Phenanthrene	Source	2,4,6-Trichlorophenol #	Source	#N/A	Source	#N/A	Source	#N/A	Source
No. Nondetect	**0**		**0**		**0**		**0**		**0**		**0**	
Data		Not Detected 5/2/2016 (<10 ug/l) 8/1/2016 (<10 ug/l) 10/6/2016 (<10 ug/l) 2/2/2017 (<1.8 ug/l)		Not Detected 5/2/2016 (<10 ug/l) 8/1/2016 (<10 ug/l) 10/6/2016 (<10 ug/l) 2/2/2017 (<1.0 ug/l)		Not Detected 5/2/2016 (<10 ug/l) 8/1/2016 (<10 ug/l) 10/6/2016 (<10 ug/l) 2/2/2017 (<5.0 ug/l)						

PROJECTED EFFLUENT QUALITY ANALYSIS (<10 Detectable):
* all values in ug/l

	3,3'-Dichlorobenzidine #	Benzidine #	Hexachlorobenzene # @	Hexachlorobutadiene # @	Hexachlorocyclopentadiene	Nitrobenzene #	Acrylonitrile #	Acenaphthylene	Azobenzene #	Butyl benzyl phthalate	4-Chloro-3-methylphenol	Di-n-butyl phthalate	Fluoranthene	Hexachloroethane #	Pentachlorophenol #	Phenanthrene	2,4,6-Trichlorophenol #	#N/A	#N/A	#N/A
Sample Max.	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Total Samples	4	4	4	4	4	4	4	4	4	0	0	0	0	0	0	0	0	0	0	0
Multiplier	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	0	0	0	0	0	0	0	0	0	0	0
PEQ:	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**